

Prove consecutive-service-rate optimality for sojourn-time partitioning under moment monotonicity

ChatGPT (gpt-5.6-sol)*

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Verifier verdict: APPROVED — progress score 3/3 (full solution)

Independent A.8 verifier assessment; pipeline details in the attribution footnote.

Abstract

We consider parallel FCFS queues sharing one unit of divisible service capacity. The decision maker jointly chooses the capacity allocated to each queue and a static, type-dependent randomized routing policy. The objective is the mean sojourn time. Customer types are ordered so that their reciprocal mean service requirements are strictly decreasing, while their second-moment-to-mean ratios are nondecreasing. We prove that every global optimum routes each positive-rate type deterministically. Moreover, after relabeling the active queues, each queue receives a consecutive block of customer types. The proof combines a pooling inequality, a single-crossing argument for the routing marginals, and a second-order perturbation that excludes splitting at block boundaries.

1 The open problem

This paper addresses an open problem posed by Shengyu Cao, Simai He, Zizhuo Wang, Yifan Feng in *Optimal Partition for a Multi-Type Queueing System* [1] (Mathematics of Operations Research, 2025), Page 34, Section 5 (Extension: Analysis of Sojourn Time), near the discussion following Theorem 8; reiterated in Section 6 (Concluding Remarks) on page 35.. The website catalogue entry is problem 74 (W4226364838_p0) of the [Open Problems in Operations Research](#) collection.

In the joint partition-and-assignment optimization problem

$$\min_{\alpha, X} \tilde{W}(\alpha, X) \quad s.t. \quad \alpha_j \geq 0, \sum_{j=1}^k \alpha_j = 1, x_{ij} \in [0, 1], \sum_{j=1}^k x_{ij} = 1 \quad \forall i, \alpha_j > \sum_{i=1}^n \lambda_i x_{ij} / \mu_i \quad \forall j \text{ with } \alpha_j > 0,$$

prove (or disprove) that under the moment monotonicity condition $\mathbb{E}[S_i^2]/\mathbb{E}[S_i]$ non-decreasing in $\mathbb{E}[S_i]$ and with $\mu_1 > \dots > \mu_n$, there exists an optimal solution (α^, X^*)*

*Generated by the `base_solver_non_interactive` (MOR sweep) pipeline (Codex CLI, non-interactive; reasoning effort `ultra`; stages A.6 solve $\rightarrow$ A.8 verify $\rightarrow$ A.9 write-up; run `run_20260821_195329_80c8a0ed, sweep-audit-approved`). Independent verifier assessment: APPROVED with progress score 3/3 (full solution). Maintained by the AI in Mathematics & Operations Research project.

in which customers are assigned to queues in consecutive blocks by service rate, i.e., there exist indices

$$1 = i_0 \leq i_1 \leq \dots \leq i_{k-1} \leq i_k = n + 1$$

such that for each queue $j \in \{1, \dots, k\}$, X^* assigns every type $i \in \{i_{j-1}, \dots, i_j - 1\}$ to queue j with probability 1 (and assigns types outside this block to queue j with probability 0).

The remainder of this document is the solution produced by the model, reproduced verbatim from the pipeline output.

2 Model

There are $n \geq 1$ customer types and $q \geq 1$ available queues. Type i arrivals form an independent Poisson process of rate

$$\lambda_i > 0.$$

A type i customer has a nonnegative service requirement S_i , measured in units of work, with

$$m_i := \mathbb{E}[S_i] \in (0, \infty), \quad \nu_i := \mathbb{E}[S_i^2] \in (0, \infty).$$

Thus, ν_i denotes the raw second moment, not the variance.

Let

$$\Lambda := \sum_{i=1}^n \lambda_i, \quad \rho := \sum_{i=1}^n \lambda_i m_i.$$

Total service capacity is normalized to one. A capacity allocation is a vector

$$\alpha = (\alpha_1, \dots, \alpha_q) \in \mathbb{R}_+^q, \quad \sum_{j=1}^q \alpha_j = 1.$$

Queue j works at speed α_j , so a job with service requirement S_i has actual service time S_i/α_j when assigned to that queue.

Routing is static and type dependent. A type i arrival is sent to queue j with probability x_{ij} , where

$$x_{ij} \geq 0, \quad \sum_{j=1}^q x_{ij} = 1.$$

Routing decisions are independent of the queue states and of the realized service requirements.

We adopt the natural convention that no traffic may be routed to a zero-capacity queue. An empty zero-capacity queue contributes zero to the objective.

For queue j , define

$$A_j := \sum_{i=1}^n \lambda_i m_i x_{ij}, \tag{1}$$

$$B_j := \sum_{i=1}^n \lambda_i \nu_i x_{ij}, \tag{2}$$

$$C_j := \sum_{i=1}^n \lambda_i x_{ij}. \tag{3}$$

Here A_j is the workload arrival rate at unit speed, C_j is the customer arrival rate, and B_j is the arrival-rate-weighted raw second moment.

A queue is called *active* if it has positive capacity and carries positive traffic. Every active queue must satisfy

$$A_j < \alpha_j.$$

We write

$$d_j := \alpha_j - A_j$$

for its capacity slack.

2.1 The sojourn-time objective

Queue j receives a Poisson stream of rate C_j . Conditional on being routed to queue j , the second moment of the unscaled service requirement is B_j/C_j . Its utilization is A_j/α_j . The Pollaczek–Khinchine formula therefore gives the mean waiting time at queue j as

$$W_j = \frac{B_j}{2\alpha_j(\alpha_j - A_j)}.$$

The fraction of all arrivals routed to queue j is C_j/Λ , while the arrival-rate-weighted service-time contribution at that queue is A_j/α_j . Hence the mean sojourn time is

$$\mathcal{T}(\alpha, x) = \frac{1}{\Lambda} \sum_{j=1}^q \left[\frac{C_j B_j}{2\alpha_j(\alpha_j - A_j)} + \frac{A_j}{\alpha_j} \right]. \quad (1)$$

Since $1/\Lambda$ is a positive constant, it is equivalent to minimize the unnormalized objective

$$\mathcal{F}(\alpha, x) := \sum_{j=1}^q \left[\frac{C_j B_j}{2\alpha_j(\alpha_j - A_j)} + \frac{A_j}{\alpha_j} \right]. \quad (2)$$

3 Existence and boundary conventions

We assume throughout that

$$\rho < 1. \quad (3)$$

This condition is necessary for feasibility, since every stable allocation satisfies

$$\rho = \sum_j A_j < \sum_{j:A_j>0} \alpha_j \leq 1.$$

It is also sufficient: one may route all traffic to a single queue with capacity one.

For completeness, define the extended queue cost

$$\bar{f}(A, B, C, \alpha) = \begin{cases} \frac{CB}{2\alpha(\alpha - A)} + \frac{A}{\alpha}, & \alpha > A \geq 0, \\ 0, & \alpha = A = B = C = 0, \\ +\infty, & \text{otherwise.} \end{cases} \quad (4)$$

Proposition 1 (Existence). *Under (3), the optimization problem admits a global minimizer.*

Proof. The capacity simplex and the product of the routing simplices form a compact set. The aggregate quantities A_j, B_j, C_j depend continuously on the routing variables.

The extended cost in (4) is lower semicontinuous. It is continuous in the strictly stable region. If a loaded queue approaches $A = \alpha > 0$, then $B, C > 0$ and its waiting term diverges to $+\infty$. At the empty point $A = B = C = \alpha = 0$, the cost is zero and all nearby costs are nonnegative.

The sum of these extended costs is therefore lower semicontinuous on a compact set and attains its minimum. Full pooling supplies a finite feasible point because $\rho < 1$, so the minimum is finite. Any finite minimizer sends no traffic to a zero-capacity queue and is strictly stable at every loaded queue. $\square$

Lemma 1 (No capacity is wasted). *At every optimum, a queue has positive capacity if and only if it carries positive traffic.*

Proof. Finiteness already implies that a zero-capacity queue is empty. Conversely, suppose an empty queue has positive capacity. There is some other active queue, since $\Lambda > 0$. Holding its traffic fixed, the derivative of an active queue's contribution with respect to its capacity is

$$\frac{\partial}{\partial \alpha} \left[\frac{CB}{2\alpha(\alpha - A)} + \frac{A}{\alpha} \right] = -\frac{CB(2\alpha - A)}{2\alpha^2(\alpha - A)^2} - \frac{A}{\alpha^2} < 0. \quad (5)$$

Transferring a sufficiently small amount of capacity from the empty queue to the active queue therefore strictly decreases the objective. $\square$

4 Ordering assumption and workload coordinates

Define

$$\mu_i := \frac{1}{m_i}, \quad \theta_i := \frac{\nu_i}{m_i}. \quad (6)$$

The parameter μ_i is the reciprocal mean service requirement, and θ_i is the second-moment-to-mean ratio.

Relabel the types so that

$$\mu_1 > \mu_2 > \cdots > \mu_n, \quad \theta_1 \leq \theta_2 \leq \cdots \leq \theta_n. \quad (7)$$

Thus the reciprocal means are strictly decreasing, whereas the second-moment-to-mean ratios are nondecreasing. Equality among the θ_i 's is allowed.

It is convenient to express routing in workload units. Let

$$w_i := \lambda_i m_i, \quad z_{ij} := w_i x_{ij}. \quad (8)$$

Then

$$z_{ij} \geq 0, \quad \sum_j z_{ij} = w_i,$$

and

$$A_j = \sum_i z_{ij}, \quad (4)$$

$$B_j = \sum_i \theta_i z_{ij}, \quad (5)$$

$$C_j = \sum_i \mu_i z_{ij}. \quad (6)$$

5 Opposite ordering of active queues

For every active queue, define

$$t_j := \alpha_j d_j, \quad u_j := \frac{B_j}{\alpha_j d_j}, \quad v_j := \frac{C_j}{\alpha_j d_j}. \quad (9)$$

Lemma 2 (Opposite queue order). *At an optimum, any two distinct active queues j and k satisfy*

$$(u_j - u_k)(v_j - v_k) < 0. \quad (10)$$

Consequently, the active queues can be relabeled so that

$$u_1 < u_2 < \dots < u_r, \quad v_1 > v_2 > \dots > v_r. \quad (11)$$

Proof. Consider two active queues, indexed 1 and 2. The identity

$$\begin{aligned} t_1 u_1 v_1 + t_2 u_2 v_2 - \frac{(t_1 u_1 + t_2 u_2)(t_1 v_1 + t_2 v_2)}{t_1 + t_2} \\ = \frac{t_1 t_2}{t_1 + t_2} (u_1 - u_2)(v_1 - v_2) \end{aligned} \quad (12)$$

follows by direct expansion.

Suppose

$$(u_1 - u_2)(v_1 - v_2) \geq 0.$$

Merge the two queues by combining both their traffic and their capacities. The merged capacity and slack are

$$\alpha_1 + \alpha_2, \quad d_1 + d_2,$$

so the merged queue is stable. Moreover,

$$(\alpha_1 + \alpha_2)(d_1 + d_2) = t_1 + t_2 + \alpha_1 d_2 + \alpha_2 d_1 \quad (7)$$

$$> t_1 + t_2. \quad (13)$$

Since $t_j u_j = B_j$ and $t_j v_j = C_j$, equations (12)–(13) imply

$$\frac{B_1 C_1}{t_1} + \frac{B_2 C_2}{t_2} \geq \frac{(B_1 + B_2)(C_1 + C_2)}{t_1 + t_2} \quad (8)$$

$$> \frac{(B_1 + B_2)(C_1 + C_2)}{(\alpha_1 + \alpha_2)(d_1 + d_2)}. \quad (14)$$

Thus the merged waiting contribution is strictly smaller.

The service contribution also decreases strictly:

$$\frac{A_1}{\alpha_1} + \frac{A_2}{\alpha_2} > \frac{A_1 + A_2}{\alpha_1 + \alpha_2}. \quad (15)$$

The merging operation is feasible and strictly improves the objective, contradicting optimality. Therefore (10) holds.

It follows in particular that all u_j 's and all v_j 's are distinct. Ordering the queues by increasing u_j then orders them by strictly decreasing v_j . $\square$

6 Single crossing and ordered supports

For an active queue j , direct differentiation of twice its queue cost gives

$$\begin{aligned} H_j(i) &:= 2 \frac{\partial}{\partial z_{ij}} \left[\frac{C_j B_j}{2\alpha_j d_j} + \frac{A_j}{\alpha_j} \right] \\ &= u_j \mu_i + v_j \theta_i + \frac{B_j C_j}{\alpha_j d_j^2} + \frac{2}{\alpha_j}. \end{aligned} \quad (16)$$

Equivalently,

$$\frac{1}{\lambda_i m_i} \frac{\partial(2\mathcal{F})}{\partial x_{ij}} = H_j(i).$$

If $z_{ij} > 0$, an infinitesimal amount of type i workload can be moved from queue j to any other active queue. The routing KKT conditions therefore give

$$z_{ij} > 0 \implies H_j(i) \leq H_k(i) \quad \text{for every active } k. \quad (17)$$

Lemma 3 (No crossing). *Order the active queues as in (11). If $a < b$ and $i < k$, then*

$$z_{ib} z_{ka} = 0. \quad (18)$$

Consequently, the supports of the active queues are ordered intervals, with possible overlap only at a boundary type.

Proof. For $a < b$, define

$$\Delta_{ab}(i) := H_a(i) - H_b(i).$$

The terms in (16) that do not depend on i contribute only a constant. Hence, for $i < k$,

$$\Delta_{ab}(k) - \Delta_{ab}(i) = (u_a - u_b)(\mu_k - \mu_i) + (v_a - v_b)(\theta_k - \theta_i). \quad (19)$$

By (7) and (11),

$$u_a - u_b < 0, \quad \mu_k - \mu_i < 0,$$

so the first term in (19) is strictly positive. Also,

$$v_a - v_b > 0, \quad \theta_k - \theta_i \geq 0,$$

so the second term is nonnegative. Therefore

$$\Delta_{ab}(k) > \Delta_{ab}(i) \quad (i < k). \quad (20)$$

Suppose $z_{ib} > 0$ and $z_{ka} > 0$ for some $i < k$. The KKT conditions (17) give

$$H_b(i) \leq H_a(i), \quad H_a(k) \leq H_b(k).$$

Equivalently,

$$\Delta_{ab}(i) \geq 0, \quad \Delta_{ab}(k) \leq 0,$$

contradicting (20). This proves (18).

Let

$$I_j := \{i : z_{ij} > 0\}.$$

The no-crossing property implies that, whenever $a < b$,

$$\max I_a \leq \min I_b. \quad (21)$$

It also implies that each I_j is an interval. Indeed, suppose $i < k < \ell$, with $i, \ell \in I_a$. Since type k must be routed somewhere, take b with $k \in I_b$. If $b < a$, then $k > i$ contradicts the order between I_b and I_a . If $b > a$, then $\ell > k$ contradicts the order between I_a and I_b . Therefore $b = a$.

Finally, two ordered supports can intersect in at most one index. Thus the only remaining possible randomization is the splitting of a boundary type between two or more queues. $\square$

7 Excluding split boundary types

Lemma 4 (No splitting). *At an optimum, every positive-rate type is routed to exactly one active queue.*

Proof. Suppose, to the contrary, that type h is split between two active queues $p < q$. By the preceding lemma, every type in queue p has index at most h , and every type in queue q has index at least h .

For one queue, suppress its index and write

$$f(A, B, C, \alpha) = \frac{CB}{2\alpha(\alpha - A)} + \frac{A}{\alpha}, \quad d = \alpha - A.$$

Adding one unit of type h workload changes the queue aggregates in the direction

$$(dA, dB, dC) = (1, \theta_h, \mu_h).$$

Direct differentiation gives

$$\frac{\partial f}{\partial z_h} = \frac{1}{\alpha} \left[1 + \frac{\mu_h B + \theta_h C}{2d} + \frac{BC}{2d^2} \right] =: \frac{J}{\alpha}, \quad (22)$$

and

$$-\frac{\partial f}{\partial \alpha} = \frac{1}{\alpha} \left[\frac{A}{\alpha} + \frac{BC(\alpha + d)}{2\alpha d^2} \right] =: \frac{D}{\alpha}. \quad (23)$$

Both J and D are strictly positive.

Because type h is used by both queues, routing stationarity gives

$$\frac{J_p}{\alpha_p} = \frac{J_q}{\alpha_q}. \quad (24)$$

Because both capacities are positive and the total capacity is fixed, capacity stationarity gives

$$\frac{D_p}{\alpha_p} = \frac{D_q}{\alpha_q}. \quad (25)$$

It follows that

$$\frac{J_p}{D_p} = \frac{J_q}{D_q} =: R > 0. \quad (26)$$

Consider transferring type h workload and capacity in the ratio $1 : R$:

$$\begin{aligned} z_{hp} &\mapsto z_{hp} - t, & \alpha_p &\mapsto \alpha_p - Rt, \\ z_{hq} &\mapsto z_{hq} + t, & \alpha_q &\mapsto \alpha_q + Rt. \end{aligned} \quad (27)$$

For sufficiently small positive or negative t , this is a two-sided feasible perturbation.

For each affected queue, define

$$\varphi_j(t) := f(A_j + t, B_j + \theta_h t, C_j + \mu_h t, \alpha_j + Rt). \quad (28)$$

Equations (22)–(26) give

$$\varphi'_j(0) = \frac{J_j - RD_j}{\alpha_j} = 0. \quad (29)$$

The objective along (27) is

$$G(t) := \varphi_p(-t) + \varphi_q(t),$$

and therefore $G'(0) = 0$.

We now calculate the second derivative. For either affected queue, introduce the dimensionless quantities

$$s := \frac{A}{d}, \quad \beta := \frac{B}{\theta_h A}, \quad \gamma := \frac{C}{\mu_h A}, \quad \kappa := \mu_h \theta_h = \frac{\nu_h}{m_h^2}. \quad (30)$$

In these variables,

$$J = 1 + \frac{\kappa}{2} [s(\beta + \gamma) + \beta\gamma s^2], \quad (31)$$

$$D = \frac{s}{2(1+s)} [2 + \kappa\beta\gamma s(s+2)]. \quad (32)$$

Consequently,

$$R = \frac{(1+s) [2 + \kappa s(\beta + \gamma) + \kappa\beta\gamma s^2]}{s [2 + \kappa\beta\gamma s(s+2)]}. \quad (33)$$

Substituting (33) into the second derivative of (28) yields

$$\varphi''_j(0) = \frac{\kappa \Phi_{\beta,j} \Phi_{\gamma,j}}{d_j^2 (1+s_j) [2 + \kappa\beta_j\gamma_j s_j(s_j+2)]^2}, \quad (34)$$

where

$$\Phi_\gamma = \kappa\gamma s \{ \gamma s(\beta - 1) + (\beta - \gamma) \} + 2(1 - \gamma), \quad (35)$$

$$\Phi_\beta = \kappa\beta s \{ \beta s(\gamma - 1) + (\gamma - \beta) \} + 2(1 - \beta). \quad (36)$$

These signs are determined by the ordered supports. Indeed,

$$\beta_j = \sum_i \frac{z_{ij}}{A_j} \frac{\theta_i}{\theta_h}, \quad \gamma_j = \sum_i \frac{z_{ij}}{A_j} \frac{\mu_i}{\mu_h}. \quad (37)$$

Since queue p contains only types $i \leq h$, condition (7) gives

$$\beta_p \leq 1 \leq \gamma_p. \quad (38)$$

Similarly, queue q contains only types $i \geq h$, so

$$\beta_q \geq 1 \geq \gamma_q. \quad (39)$$

If $\beta \leq 1 \leq \gamma$, then every term in

$$\Phi_\gamma = \kappa \gamma s \{ \gamma s (\beta - 1) + (\beta - \gamma) \} + 2(1 - \gamma)$$

is nonpositive, whereas every term in

$$\Phi_\beta = \kappa \beta s \{ \beta s (\gamma - 1) + (\gamma - \beta) \} + 2(1 - \beta)$$

is nonnegative. Thus

$$\Phi_\gamma \leq 0, \quad \Phi_\beta \geq 0.$$

When $\beta \geq 1 \geq \gamma$, the signs reverse. It follows from (34) that

$$\varphi_p''(0) \leq 0, \quad \varphi_q''(0) \leq 0. \quad (40)$$

Moreover, equality in either case is possible only when

$$\beta = \gamma = 1. \quad (41)$$

For example, under $\beta \leq 1 \leq \gamma$, equality in $\Phi_\gamma \leq 0$ first forces $\gamma = 1$, after which it forces $\beta = 1$. The reversed case is identical.

Unless both queues satisfy (41), equations (34) and (40) give

$$G''(0) = \varphi_p''(0) + \varphi_q''(0) < 0.$$

This contradicts the second-order necessary condition $G''(0) \geq 0$ for a local minimum along a two-sided feasible perturbation.

It remains to exclude the equality case. Since the μ_i 's are strictly ordered, the weighted-average identity

$$\gamma_j = \sum_i \frac{z_{ij}}{A_j} \frac{\mu_i}{\mu_h} = 1$$

forces an equality-case queue to contain only type h . Thus both p and q would be pure type- h queues.

For a pure type- h queue, let $r = A/\alpha \in (0, 1)$. Its objective contribution is

$$g(r) := r + \frac{\kappa r^2}{2(1-r)}. \quad (42)$$

If the two pure queues are merged, their utilization becomes

$$\bar{r} = \frac{\alpha_p r_p + \alpha_q r_q}{\alpha_p + \alpha_q},$$

which lies between r_p and r_q . Since

$$g'(r) = 1 + \frac{\kappa r(2-r)}{2(1-r)^2} > 0$$

and $g(r) > 0$ for $r > 0$,

$$g(\bar{r}) \leq \max\{g(r_p), g(r_q)\} < g(r_p) + g(r_q).$$

Merging therefore strictly improves the objective, again contradicting optimality. Hence no type can be split. $\square$

8 Main result

Theorem 1 (Deterministic consecutive-block routing). *Assume*

$$\lambda_i > 0, \quad m_i > 0, \quad \sum_i \lambda_i m_i < 1,$$

and suppose the types are ordered according to

$$\mu_1 > \mu_2 > \cdots > \mu_n, \quad \theta_1 \leq \theta_2 \leq \cdots \leq \theta_n,$$

where

$$\mu_i = \frac{1}{m_i}, \quad \theta_i = \frac{\mathbb{E}[S_i^2]}{m_i}.$$

Then every global optimum of the joint capacity-allocation and routing problem is deterministic and monotone.

More precisely, after discarding zero-capacity queues and relabeling the r active queues, there exist cut points

$$0 = k_0 < k_1 < \cdots < k_r = n$$

such that

$$x_{ij} = \begin{cases} 1, & k_{j-1} < i \leq k_j, \\ 0, & \text{otherwise.} \end{cases} \quad (43)$$

Thus the preimage of every active queue is a consecutive block of service-type indices.

Proof. Existence follows from the extended-value compactness argument.

By the opposite-order lemma, the active queues admit a strict order by increasing u_j , equivalently by decreasing v_j . The single-crossing lemma then shows that their supports are ordered intervals, possibly sharing boundary types. The no-splitting lemma eliminates every such overlap. Since all $\lambda_i > 0$, every type must consequently be routed wholly to exactly one queue.

The resulting nonempty, disjoint, ordered intervals cover $\{1, \dots, n\}$, and hence are described by cut points $0 = k_0 < k_1 < \cdots < k_r = n$. Finally, every inactive queue has zero capacity by the no-wasted-capacity lemma. $\square$

9 Scope and boundary qualifications

Remark (Zero-arrival types). The assumption $\lambda_i > 0$ is needed for the statement about *every* optimum. If a nominal type has $\lambda_i = 0$, its routing probabilities do not affect the objective and may be arbitrary. After deleting such types, the theorem applies to all remaining types. A deterministic optimal representative still exists when zero-arrival types are retained.

Remark (Raw second moments). The parameter ν_i used throughout is $\mathbb{E}[S_i^2]$. Replacing it by $\text{Var}(S_i)$ would change B_j , the Pollaczek–Khinchine term, and the ordering condition.

Remark (Zero-capacity convention). An active queue approaching $A_j = \alpha_j$ has infinite waiting cost. An empty queue with

$$A_j = B_j = C_j = \alpha_j = 0$$

instead contributes zero. These two boundary cases must be treated separately. In particular, the displayed optimization constraints must be interpreted as prohibiting traffic to a zero-capacity queue.

Remark (Strictness). Strict ordering of the μ_i 's supplies the strict part of the single-crossing argument and excludes nontrivial equality cases in the no-splitting argument. Ties among the θ_i 's are permitted.

Remark (Number of queues). The number q of available queue slots is fixed in the formulation proved here. The theorem is a structural result for joint capacity partition and assignment under the sojourn-time objective. It does not analyze a separate design problem in which the number of queues is itself an optimization variable, and therefore should not be read as resolving a separate conjecture about the optimal queue count.

Remark (Policy class). The result concerns static, type-dependent routing to FCFS $M/G/1$ queues. Queue-state-dependent routing, routing based on a job's realized size, priorities, and other service disciplines need not have objective (1) and are outside the theorem.

10 Conclusion

Under the monotone moment ordering (7), randomization cannot occur in an optimal joint routing and capacity allocation. The pooling inequality first forces two normalized queue statistics to be oppositely ordered. This produces a strict single-crossing property for the routing marginals and hence ordered interval supports. A capacity-adjusted second-order workload transfer then excludes the only remaining possibility for randomization, namely a type split at the boundary of two intervals. Consequently, every optimum assigns the customer types deterministically to consecutive blocks.

References

- [1] Shengyu Cao, Simai He, Zizhuo Wang, Yifan Feng. *Optimal Partition for a Multi-Type Queueing System*. Mathematics of Operations Research, 2025. <https://doi.org/10.1287/moor.2023.0035>